

Finding Hidden Structure in Complex Systems

Seeking:
Quant Research Internship
Summer 2026 / 2027QR Code
Portfolio &
GitHub

From Differential Geometry to Quantitative Finance

Geometric ML for Regime Detection

QCML Framework · Independent Research · Solo-authored 34-page paper

Problem

Market crashes cause catastrophic losses. Can we detect regime shifts *without labeled crisis data*?

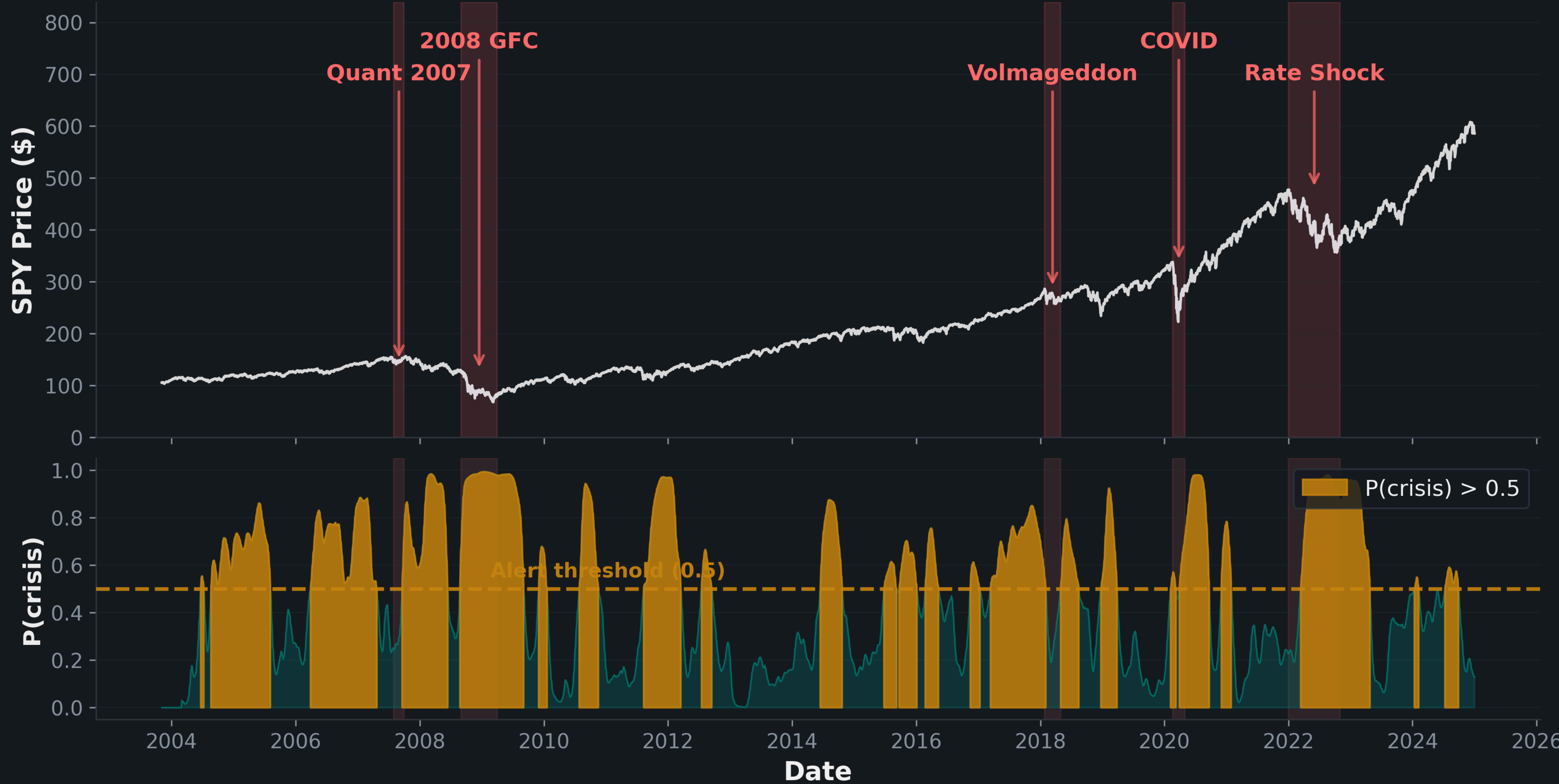
Approach

- Embed financial time series into a **geometric space** (projective Hilbert space) using spectral metric learning
- Extract **3 geometric observables** that measure how the data manifold deforms during stress
- Fully unsupervised** — no crisis labels, no regime annotations, no look-ahead

Note: "QCML" is a historical name — the math is differential geometry and spectral theory, not quantum computing.

d = 0.68QFI vs calm periods
(unsupervised, no labels)**12 / 12**Crises detected by
best geometric observable**0.26s**Multi-Lag Fidelity
faster than RF (1.07s)

Geometric Signal Detection Across Major Financial Crises (2005-2024)



Key Results

- Unsupervised** QFI ($d = 0.68$) outperforms **supervised** Random Forest ($d = 0.24$) — no crisis labels needed
- CUSUM ($d = 0.87$) strongest overall; geometric methods are **competitive and complementary**
- Each observable captures different crisis types — QFI on novel mechanisms, Berry on structural transitions
- Multi-asset generalization**: tested across 5 ETFs (SPY, QQQ, IWM, EFA, DIA) $\times$ 4 major crises

Python PyTorch NumPy/SciPy scikit-learn Optuna Polygon.io LaTeX

Production Quantitative Research

Quanta Ventures Fund · Systematic equity strategies

Problem

How do you combine 133 uncorrelated ML signals into a single portfolio that survives regime changes and transaction costs?

Approach

- 5-sleeve architecture** aggregating 133 signals across XGBoost, Ridge, and mean-reversion
- Two-stage volatility targeting with **VIX/VVIX regime diagnostics**
- Walk-forward validation with 30-day embargo and 11-test robustness suite

2.92Sharpe Ratio
(true OOS, 2022–2025)**5.02**Calmar Ratio
(return / max drawdown)**133**Signals across
5 strategy sleeves

System Architecture

133 Signals

XGB / Ridge / MR

→

5 Sleeves

Cov-weighted

→

Risk Engine

VIX/VVIX

→

Portfolio

Dynamic sizing

Signal Types

Momentum

XGBoost cross-sectional signals

Mean Reversion

Z-score convergence strategies

Statistical

Ridge regression factor models

Risk Budget

Covariance-weighted allocation

Key Results

- Sharpe **2.92** and Calmar **5.02** on true out-of-sample data (2022–2025)
- 133 signals** with covariance-based risk budgeting across 5 sleeves
- Passed **11-test validation suite** (parameter stability, bootstrap, walk-forward)
- Dynamic regime-aware leverage reduced peak drawdown via VIX/VVIX classification

Risk Management

- VIX/VVIX regime classification for **dynamic leverage adjustment**
- 30-day embargo prevents lookahead bias contamination
- Heston stochastic vol for tail-risk monitoring

Python XGBoost Optuna Polygon.io PostgreSQL FRED API

Technical Skills

Python (NumPy, Pandas, PyTorch) C++ (pybind11, FFT)

SQL (PostgreSQL) TypeScript / React

Mathematica Git / CI/CD LaTeX

Quantitative Methods

Stochastic Calculus Monte Carlo

Options Pricing (BS, Heston)

Bayesian Inference Time Series

Walk-Forward Validation

Risk Metrics (Sharpe, VaR)

Communication

CS Teaching Assistant Mentoring 60+ students in intro CS

Independent Researcher Solo-authored 34-page paper, 3 theorems

Technical Communicator Translating complex math for diverse audiences

Honors & Experience

Incoming Quant Research Intern

Steadfast Financial LP — Summer 2026

Vol Arb Platform

Real-time C++/Python/React, 40x latency reduction

Carpe Diem Endowed Scholar

Pitzer College academic distinction

Harvey Mudd Putnam Math

Competitive mathematics

Graduate-Level Coursework

Math Finance, Stochastic Calc (sophomore)